

Circuit Theory

Chapter 7

First-Order Circuits

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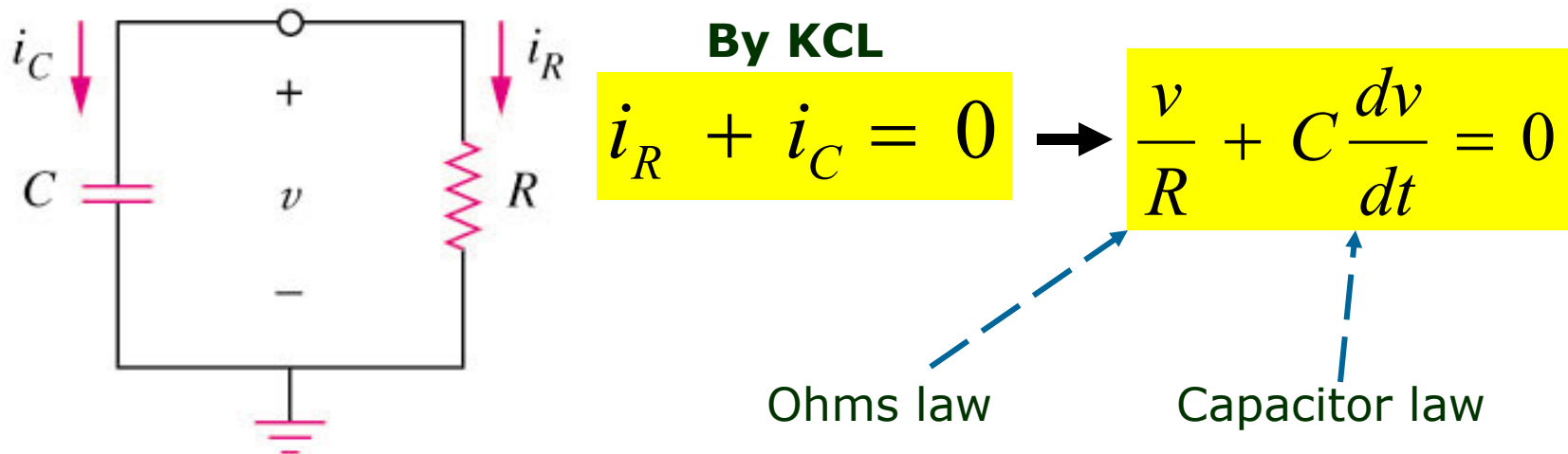
First-Order Circuits

Chapter 7

- 7.1 The Source-Free RC Circuit
- 7.2 The Source-Free RL Circuit
- 7.3 Unit-step Function
- 7.4 Step Response of an RC Circuit
- 7.5 Step Response of an RL Circuit

7.1 The Source-Free RC Circuit (1)

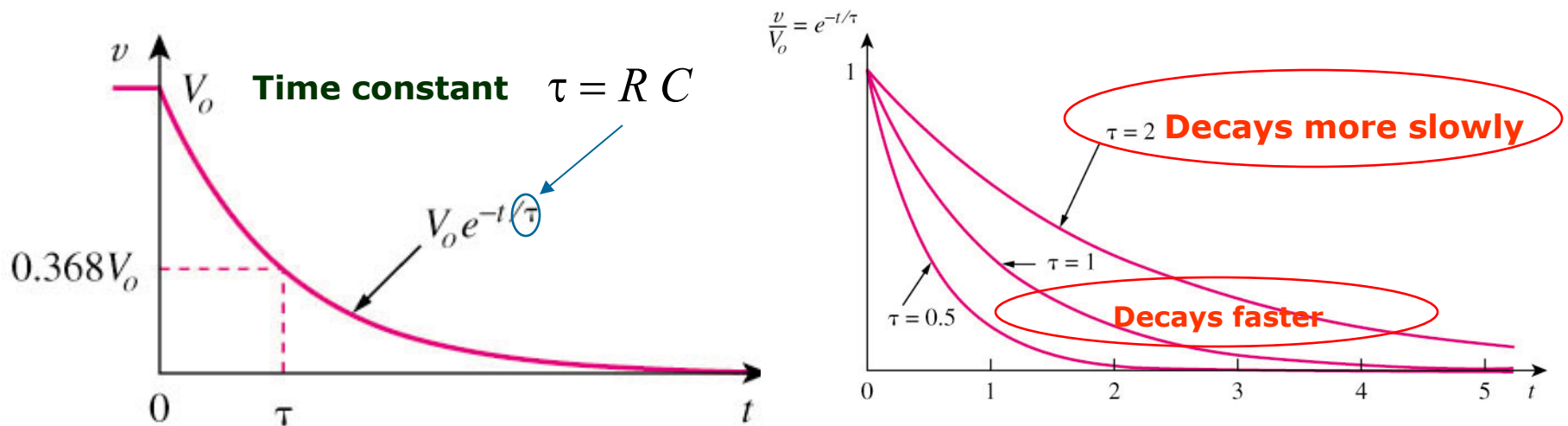
- A **first-order circuit** is characterized by a first-order differential equation.



- Apply Kirchhoff's laws to purely resistive circuit results in algebraic equations.
- Apply the laws to RC and RL circuits produces differential equations.

7.1 The Source-Free RC Circuit (2)

- The **natural response** of a circuit refers to the behavior (in terms of voltages and currents) of the circuit itself, with no external sources of excitation.

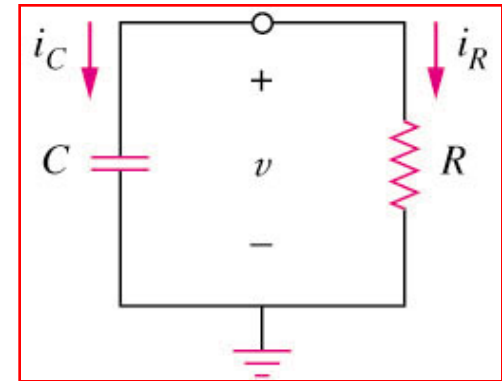


- The **time constant** τ of a circuit is the time required for the response to decay by a factor of $1/e$ or 36.8% of its initial value.
- v decays **faster** for small τ and **slower** for large τ .

7.1 The Source-Free RC Circuit (3)

The key to working with a source-free RC circuit is finding:

$$v(t) = V_0 e^{-t/\tau} \quad \text{where} \quad \tau = R C$$



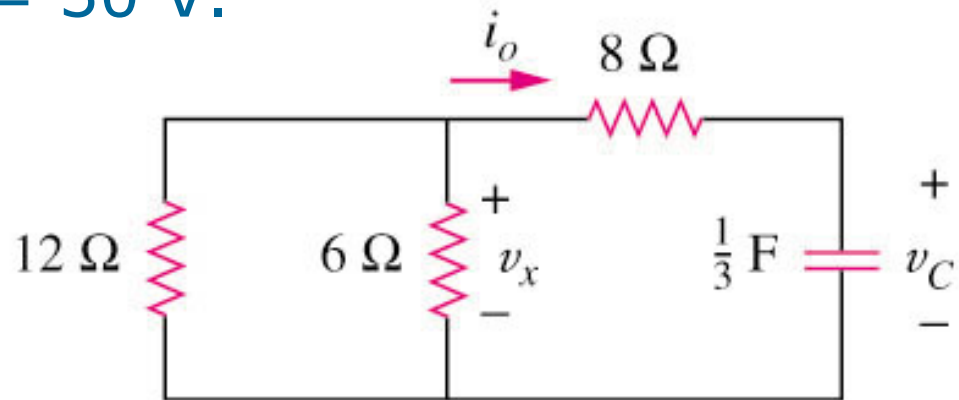
1. The initial voltage $v(0) = V_0$ across the capacitor.
2. The time constant $\tau = RC$.

7.1 The Source-Free RC Circuit (4)

Example 1

Refer to the circuit below, determine v_C , v_x , and i_o for $t \geq 0$.

Assume that $v_C(0) = 30$ V.



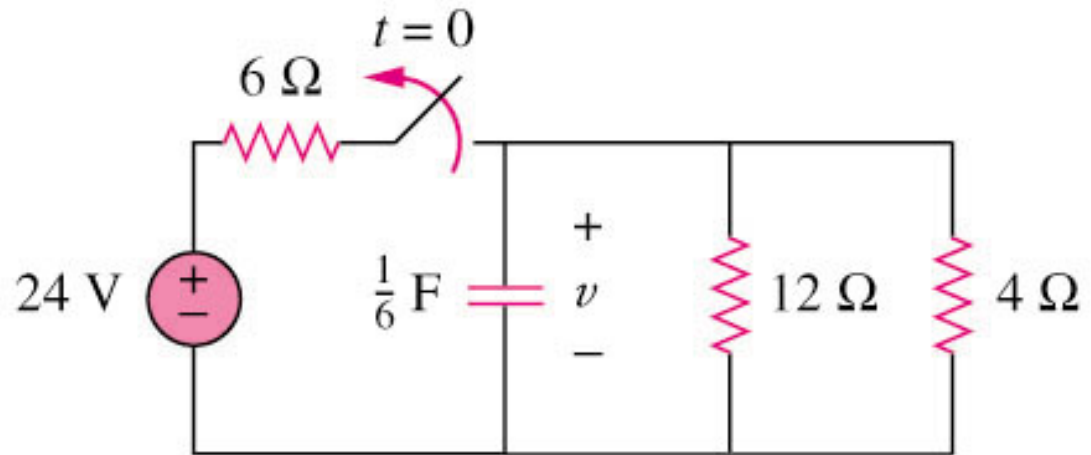
- Please refer to lecture or textbook for more detail elaboration.

Answer: $v_C = 30e^{-0.25t}$ V ; $v_x = 10e^{-0.25t}$; $i_o = -2.5e^{-0.25t}$ A

7.1 The Source-Free RC Circuit (5)

Example 2

The switch in circuit below is opened at $t = 0$, find $v(t)$ for $t \geq 0$.

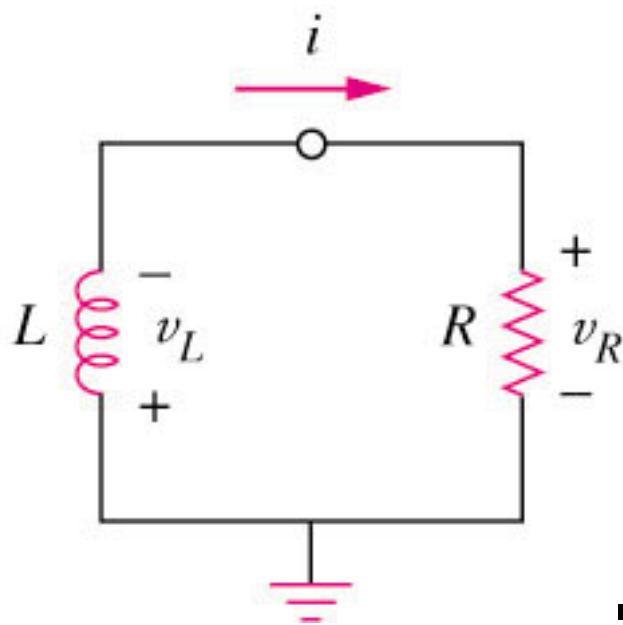


- Please refer to lecture or textbook for more detail elaboration.

Answer: $V(t) = 8e^{-2t}$ V

7.2 The Source-Free RL Circuit (1)

- A **first-order RL circuit** consists of a inductor L (or its equivalent) and a resistor (or its equivalent)



By KVL

$$v_L + v_R = 0$$

$$L \frac{di}{dt} + iR = 0$$

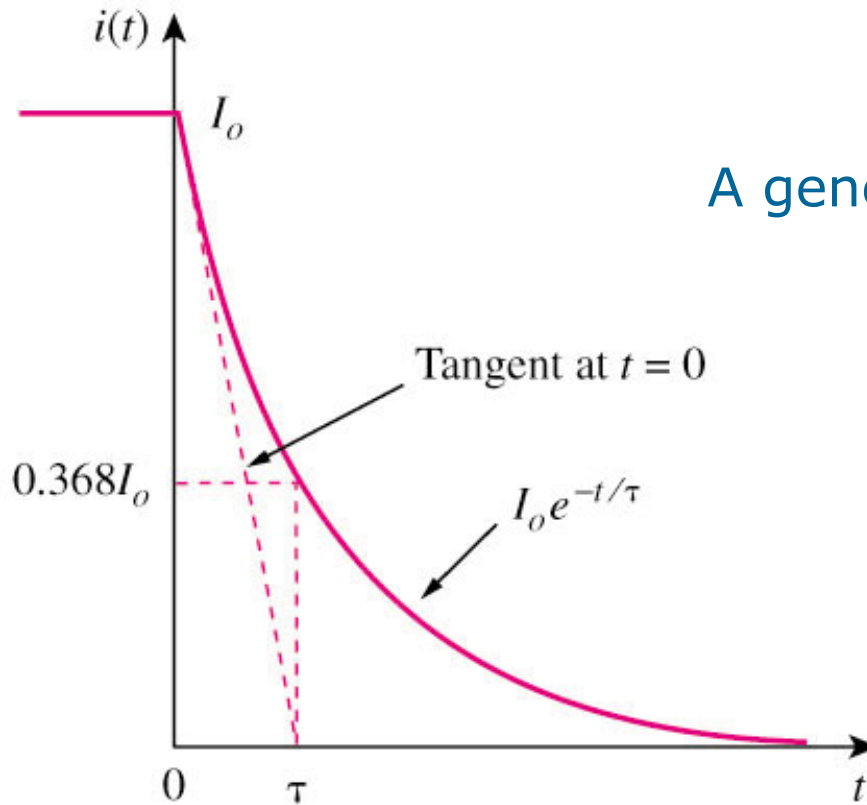
Inductors law

Ohms law

$$\frac{di}{i} = -\frac{R}{L} dt$$

$$i(t) = I_0 e^{-Rt/L}$$

7.2 The Source-Free RL Circuit (2)



A general form representing a RL

$$i(t) = I_0 e^{-t/\tau}$$

where $\tau = \frac{L}{R}$

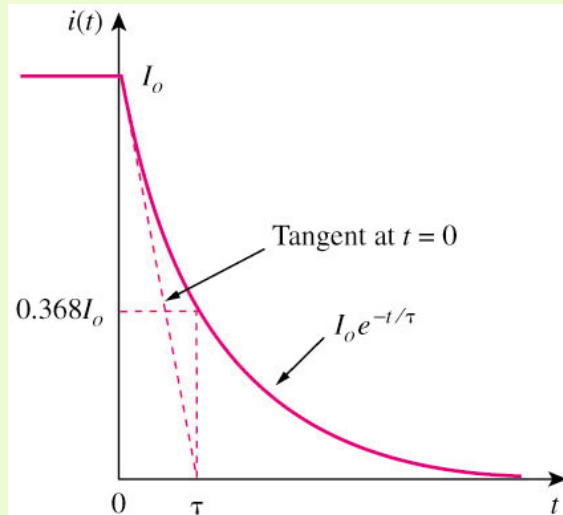
- The **time constant** τ of a circuit is the time required for the response to decay by a factor of $1/e$ or 36.8% of its initial value.
- $i(t)$ decays **faster** for small τ and **slower** for large τ .
- The general form is very similar to a RC source-free circuit.

7.2 The Source-Free RL Circuit (3)

Comparison between a RL and RC circuit

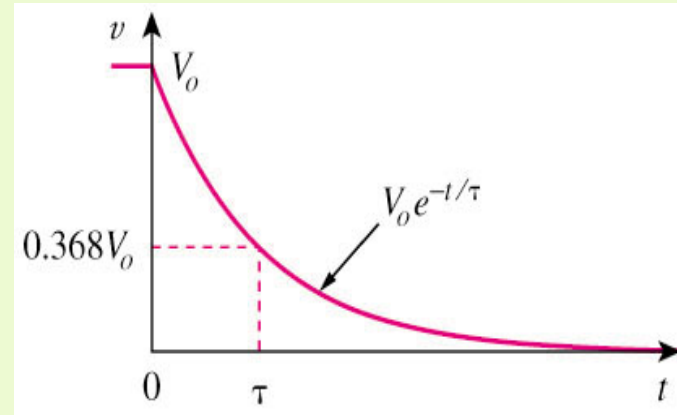
A RL source-free circuit

$$i(t) = I_0 e^{-t/\tau} \quad \text{where} \quad \tau = \frac{L}{R}$$



A RC source-free circuit

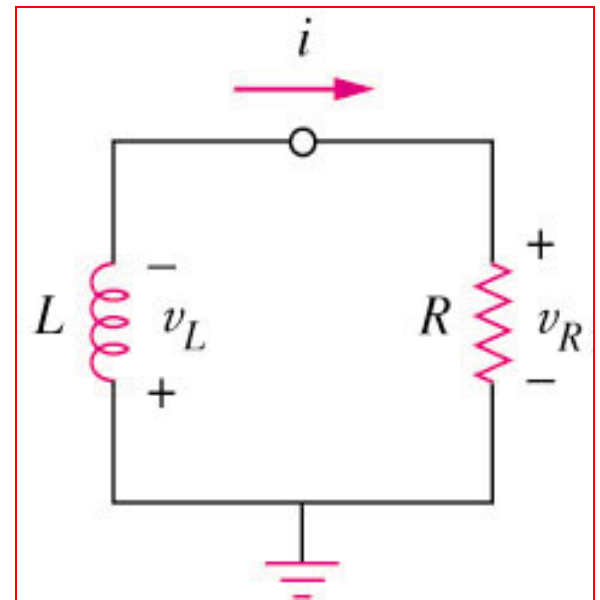
$$v(t) = V_0 e^{-t/\tau} \quad \text{where} \quad \tau = RC$$



7.2 The Source-Free RL Circuit (4)

The key to working with a source-free RL circuit is finding:

$$i(t) = I_0 e^{-t/\tau} \quad \text{where} \quad \tau = \frac{L}{R}$$



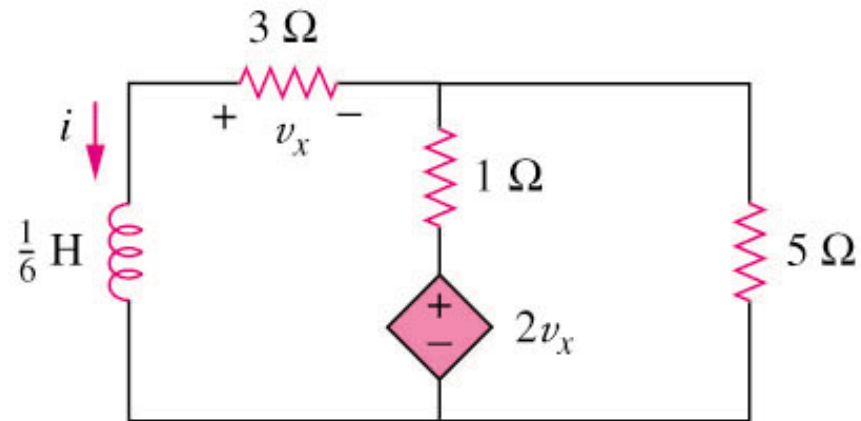
1. The initial voltage $i(0) = I_0$ through the inductor.
2. The time constant $\tau = L/R$.

7.2 The Source-Free RL Circuit (5)

Example 3

Find i and v_x in the circuit.

Assume that $i(0) = 5 \text{ A}$.



- Please refer to lecture or textbook for more detail elaboration.

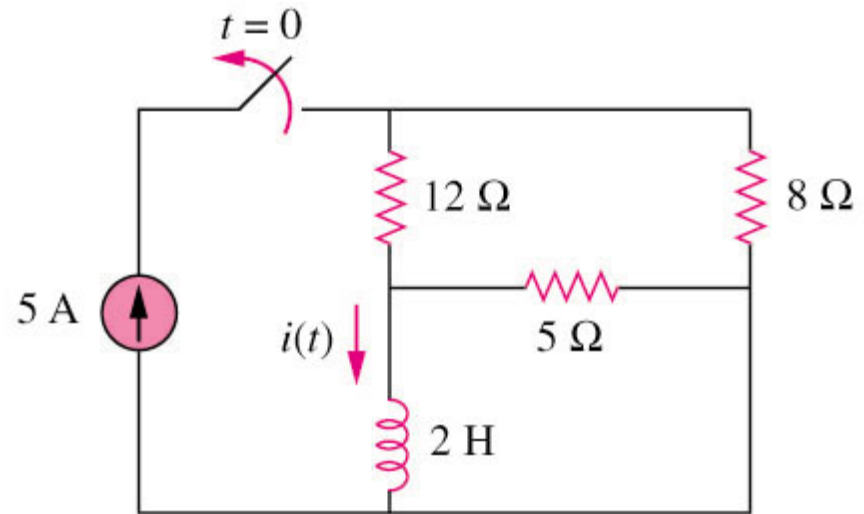
Answer: $i(t) = 5e^{-53t} \text{ A}$

$v_x(t) = -15e^{-53t} \text{ A}$

7.2 The Source-Free RL Circuit (6)

Example 4

For the circuit, find $i(t)$ for $t > 0$.



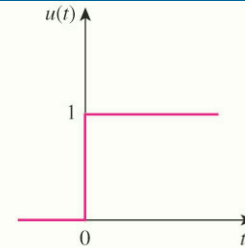
- Please refer to lecture or textbook for more detail elaboration.

Answer: $i(t) = 2e^{-2t}\text{ A}$

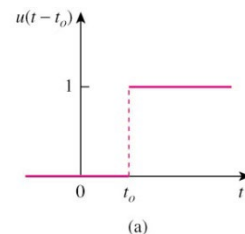
7.3 Unit-Step Function (1)

- The **unit step function** $u(t)$ is 0 for negative values of t and 1 for positive values of t .

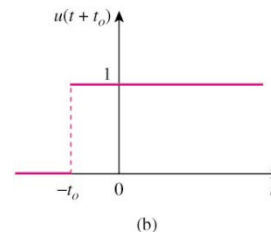
$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$



$$u(t - t_o) = \begin{cases} 0, & t < t_o \\ 1, & t > t_o \end{cases}$$



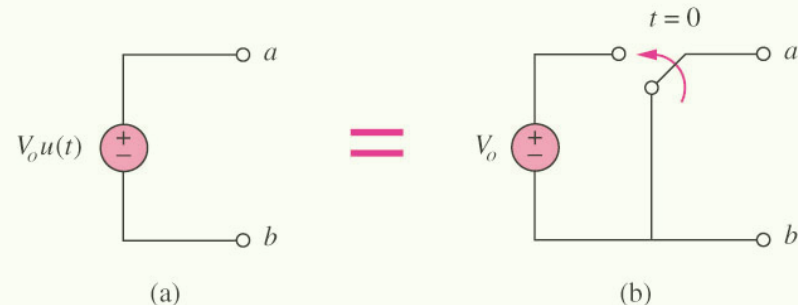
$$u(t + t_o) = \begin{cases} 0, & t < -t_o \\ 1, & t > -t_o \end{cases}$$



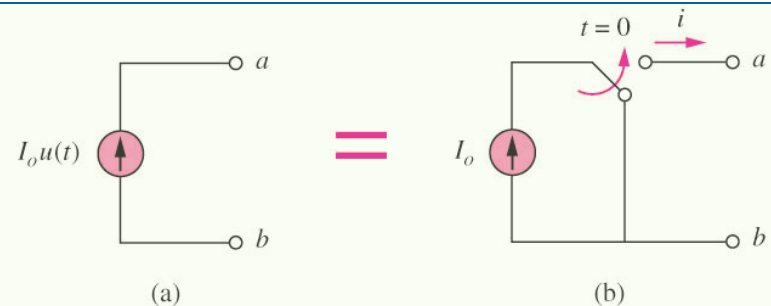
7.3 Unit-Step Function (2)

Represent an abrupt change for:

1. voltage source.

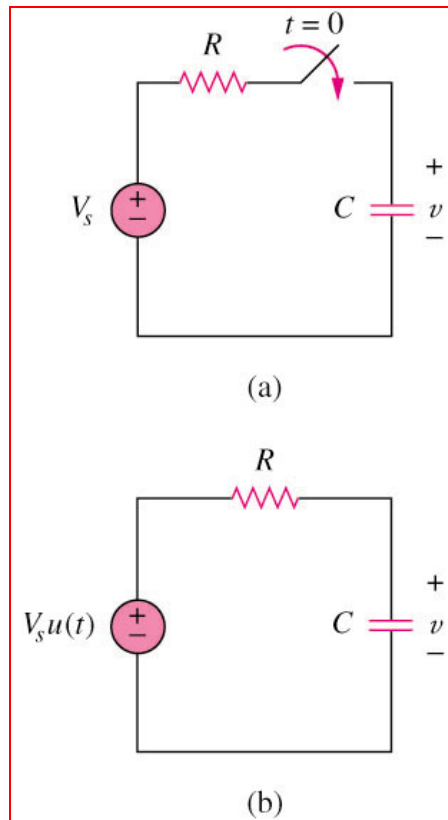


2. for current source:



7.4 The Step-Response of a RC Circuit (1)

- The **step response** of a circuit is its behavior when the excitation is the step function, which may be a voltage or a current source.



- Initial condition:**

$$v(0^-) = v(0^+) = V_0$$

- Applying KCL,

$$C \frac{dv}{dt} + \frac{v - V_s u(t)}{R} = 0$$

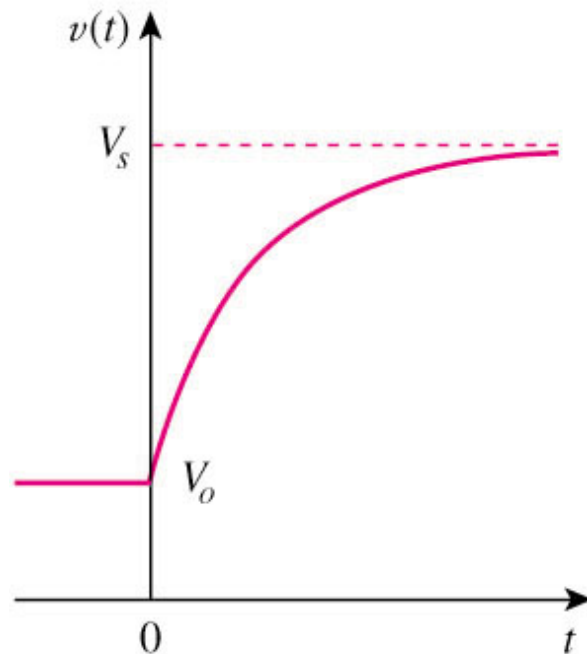
or

$$\frac{dv}{dt} = - \frac{v - V_s}{RC} u(t)$$

- Where $u(t)$ is the unit-step function

7.4 The Step-Response of a RC Circuit (2)

- Integrating both sides and considering the initial conditions, the solution of the equation is:



$$v(t) = \begin{cases} V_0 & t < 0 \\ V_s + (V_0 - V_s)e^{-t/\tau} & t > 0 \end{cases}$$

Final value
at $t \rightarrow \infty$

Initial value
at $t = 0$

Source-free
Response

Complete Response =	Natural response (stored energy)	+	Forced Response (independent source)
=	$V_0 e^{-t/\tau}$	+	$V_s (1 - e^{-t/\tau})$

7.4 The Step-Response of a RC Circuit (3)

Three steps to find out the step response of an RC circuit:

1. The initial capacitor voltage $v(0)$.
2. The final capacitor voltage $v(\infty)$ — DC voltage across C.
3. The time constant τ .

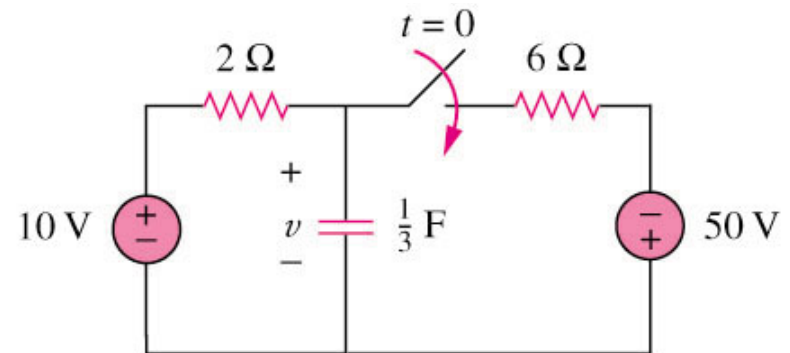
$$v(t) = v(\infty) + [v(0+) - v(\infty)]e^{-t/\tau}$$

Note: The above method is a short-cut method. You may also determine the solution by setting up the circuit formula directly using KCL, KVL, ohms law, capacitor and inductor VI laws.

7.4 The Step-Response of a RC Circuit (4)

Example 5

Find $v(t)$ for $t > 0$ in the circuit in below.
Assume the switch has been open for a long time and is closed at $t = 0$.
Calculate $v(t)$ at $t = 0.5$.

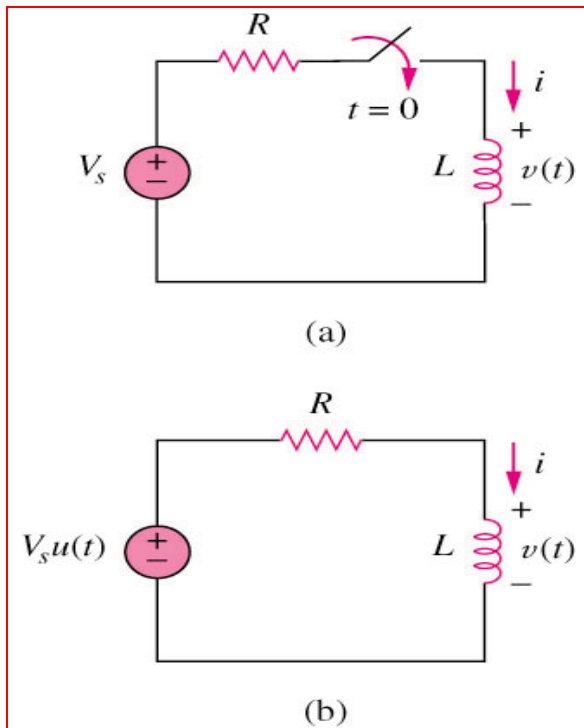


- Please refer to lecture or textbook for more detail elaboration.

Answer: $v(t) = 15e^{-2t} - 5$ and $v(0.5) = 0.5182V$

7.5 The Step-response of a RL Circuit (1)

- The **step response** of a circuit is its behavior when the excitation is the step function, which may be a voltage or a current source.



- Initial current**
 $i(0^-) = i(0^+) = I_o$
- Final inductor current**
 $i(\infty) = V_s/R$
- Time constant $\tau = L/R$

$$i(t) = \frac{V_s}{R} + \left(I_o - \frac{V_s}{R}\right) e^{-\frac{t}{\tau}} u(t)$$

7.5 The Step-Response of a RL Circuit (2)

Three steps to find out the step response of an RL circuit:

1. The initial inductor current $i(0)$ at $t = 0+$.
2. The final inductor current $i(\infty)$.
3. The time constant τ .

$$i(t) = i(\infty) + [i(0+) - i(\infty)] e^{-t/\tau}$$

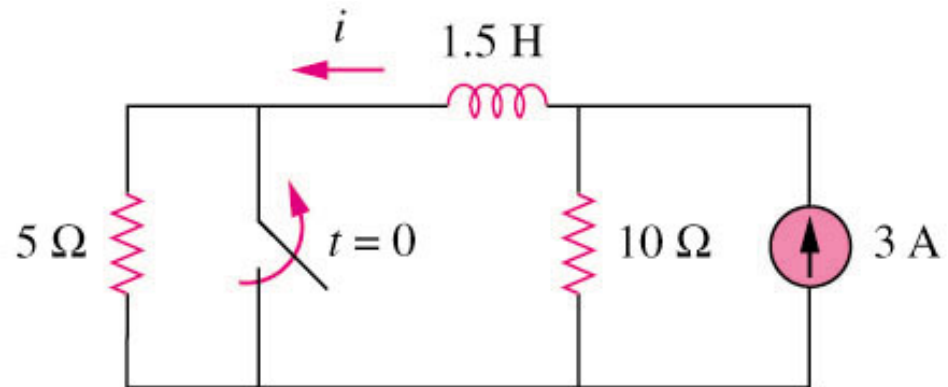
Note: The above method is a short-cut method. You may also determine the solution by setting up the circuit formula directly using KCL, KVL, ohms law, capacitor and inductor VI laws.

7.5 The Step-Response of a RL Circuit (4)

Example 6

The switch in the circuit shown below has been closed for a long time. It opens at $t = 0$.

Find $i(t)$ for $t > 0$.



- Please refer to lecture or textbook for more detail elaboration.

Answer: $i(t) = 2 + e^{-10t}$